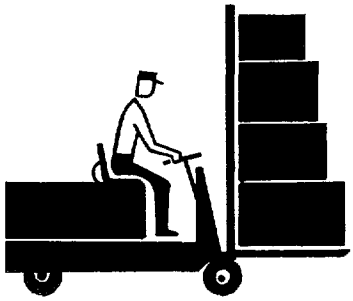
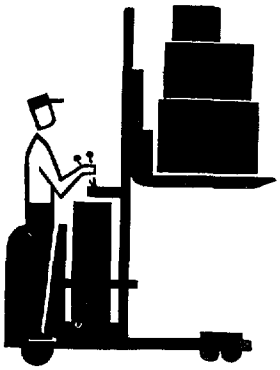
OPERATING & MAINTENANCE

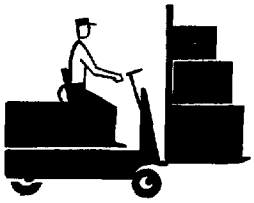
GEK—7938A

INSTRUCTIONS
IC 4483
C-290

SCR CONTROL for ELECTRIC VEHICLES







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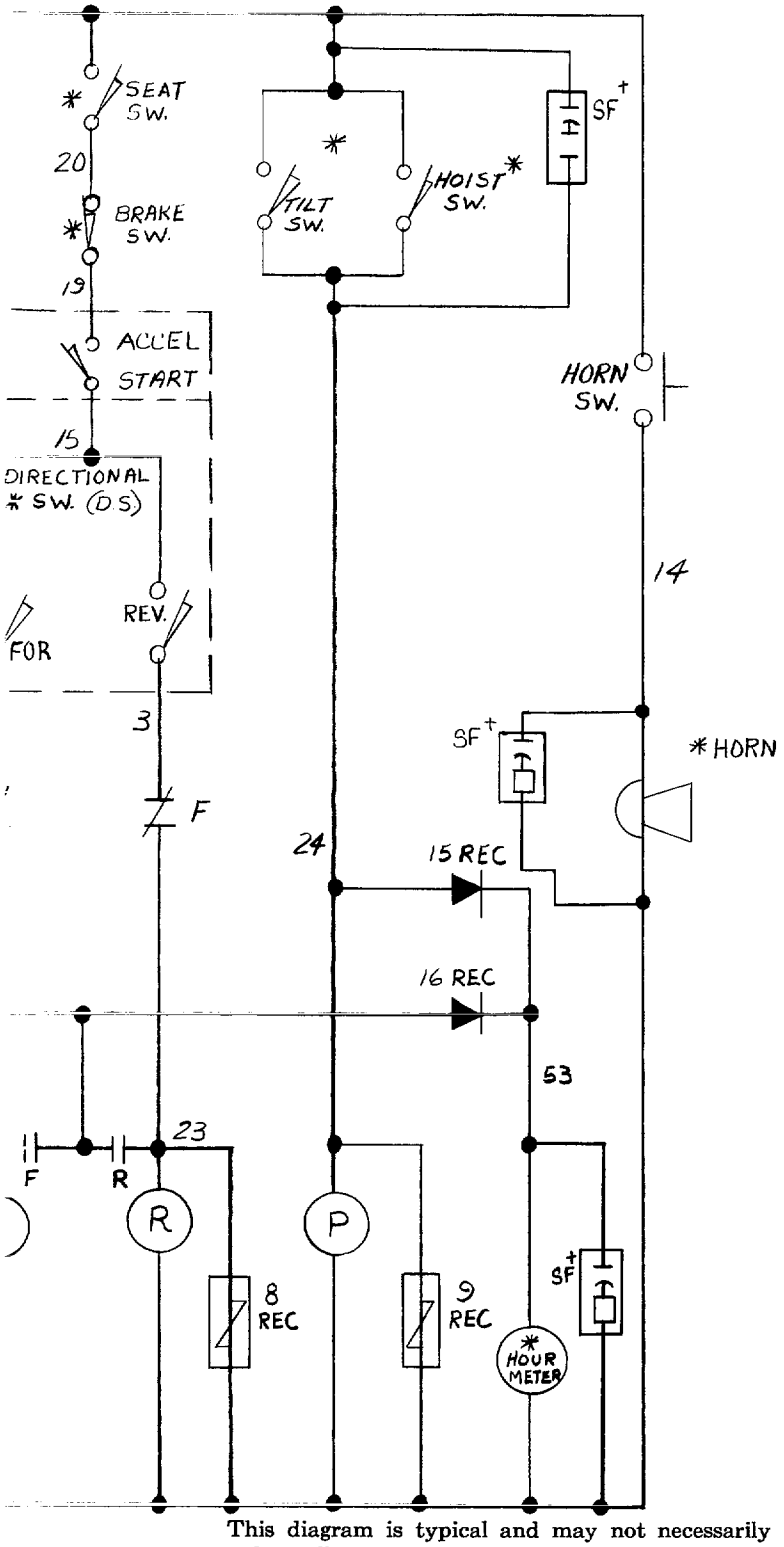
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The information contained herein is intended to assist truck users and dealers in the servicing of SCR control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance.

Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the truck manufacturer through his normal service channels, not directly to General Electric Company.

GENERAL ELECTRIC

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This diagram is typical and may not necessarily reflect all the details of every truck application. Refer to the diagram furnished with the vehicle.

SCR control.

*Furnished by Truck Mfr.

†Installed by Truck Mfr.

CIRCUIT OPERATION

(See Figure 4)

The circuit is energized by closing the key switch, the seat switch, the brake switch and moving the For. or Rev. lever to either position and then depressing the accelerator. The F or R contactor coil is now energized applying power to the drive motor circuit. Positive control power is fed through F or R interlock to an oscillator located in Card 1. The oscillator section will oscillate only when it receives both positive power through the F or R interlock and a synchronizing control signal from the anode of 1 REC. The oscillator output is fed from terminal 4 to the gate of 1 REC, the main SCR. This is the gate signal which will switch 1 REC to the conducting state. When 1 REC is conducting, current flow is from battery positive through 1 FU, drive motor, T2-T1, 1 REC and back to battery negative. The initial rising d-c current through T2-T1 induces a voltage from T4 to T3, drives T3 below battery negative, causes current to flow through 9 RES to the gate of 5 REC, turning 5 REC on. The current flowing through 5 REC and T3 to T4 charges 1C (wire 34) negative until the transformer saturates, reducing this current flow to zero, turning off 5 REC. The voltage of T3 then swings from negative to positive, causes current to flow through 8 RES to the gate of 2 REC, turning 2 REC on. 2 REC conducts, capacitor 1C discharges around the circuit composed of 1C, 1 REC, 2 REC and 1X. This discharge current opposes the battery current through 1 REC so that the resultant current is zero. With reverse voltage across 1 REC, the main SCR (1 REC) is turned off.

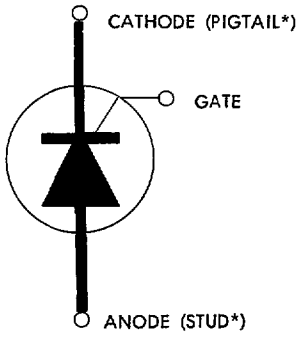
This explanation has been for one complete cycle of circuit operation. The time for the next cycle to start is determined by the time that the oscillator section takes to oscillate, which is a function of the position of the accelerator potentiometer (AS POT). Figure 5 illustrates the pulsing of current from the battery. During the off time, the energy stored in the motor by virtue of its inductance will cause current to circulate through the motor around the loop formed by 3 REC. Figure 6 shows the nature of the motor current. It should be noted that the average current measured at the motor will be greater than the average current measured at the battery. The SCR control in effect converts battery current at battery volts into a higher motor current and a lower motor volts.

CARD 1—Card 1 performs five basic control functions, and four are provided with a trimpot adjustment: (Sequence of adjustment for the trimpots is outlined in Table 6 and must be followed in order to obtain proper setting for each function.)

- CREEP-SPEED—The creep-speed trimpot adjusts the minimum speed of the truck when the accelerator is moved to the start position.
- TOP-SPEED—The top-speed trimpot adjusts the frequency of the oscillator for the main SCR thus controlling the top SCR speed of the truck prior to the time that the 1A (bypass) contactor picks up.

WHAT IS AN SCR?

Since the heart of the control is a silicon controlled rectifier (SCR), a general understanding of the characteristics of the device will be helpful. The SCR is a semiconductor rectifier used as a latching switch; i.e., it may assume either a conducting or nonconducting state (On or Off).



The SCR can be turned on by a momentary application of control current to the gate. To turn it off, it is necessary in addition to removing the turn-on signal from the gate, either to remove all power from the SCR or to apply momentary reverse voltage between cathode and anode.

* Typical of SCR as used in GE control for electric vehicles.

PHOTOS OF CONTROL

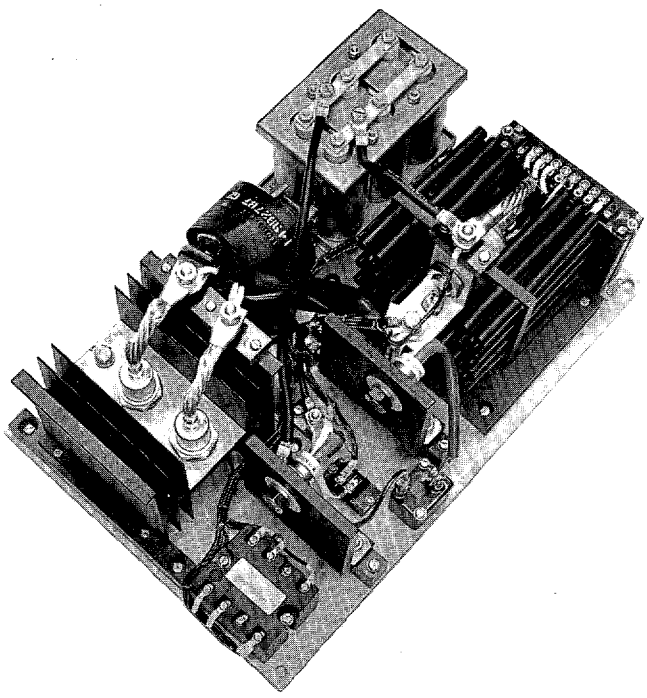


Figure 1

Typical C-290 static panel.

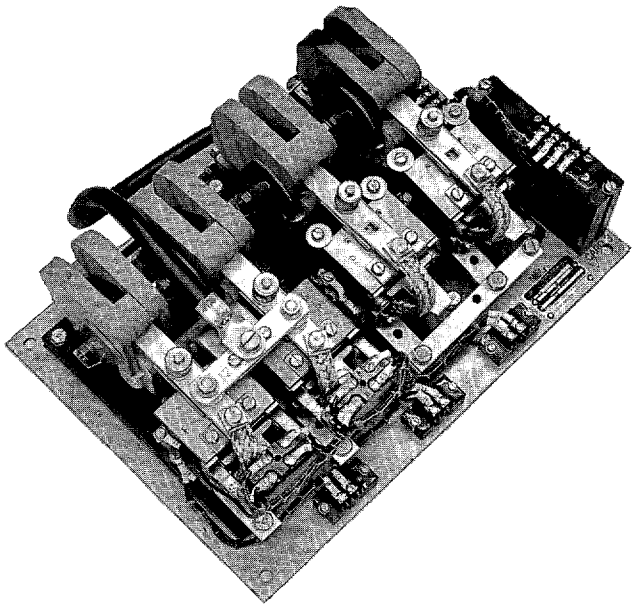


Figure 2

Typical magnetic panel consisting of forward, reverse and bypassing contactors; and static timing module.

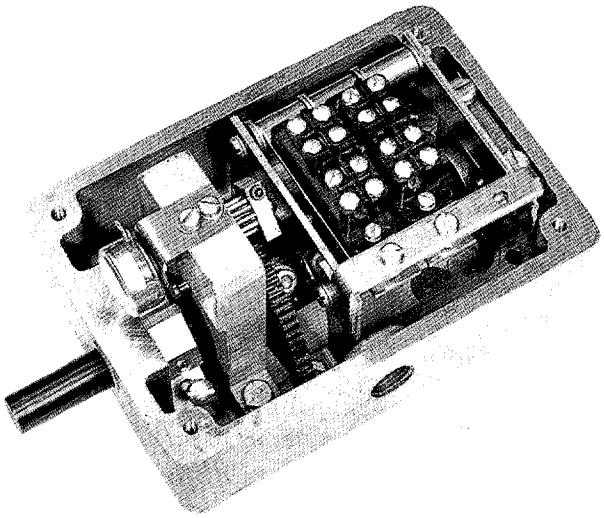


Figure 3

Typical foot-operated accelerator switch with cover removed showing speed control potentiometer and control interlocks.

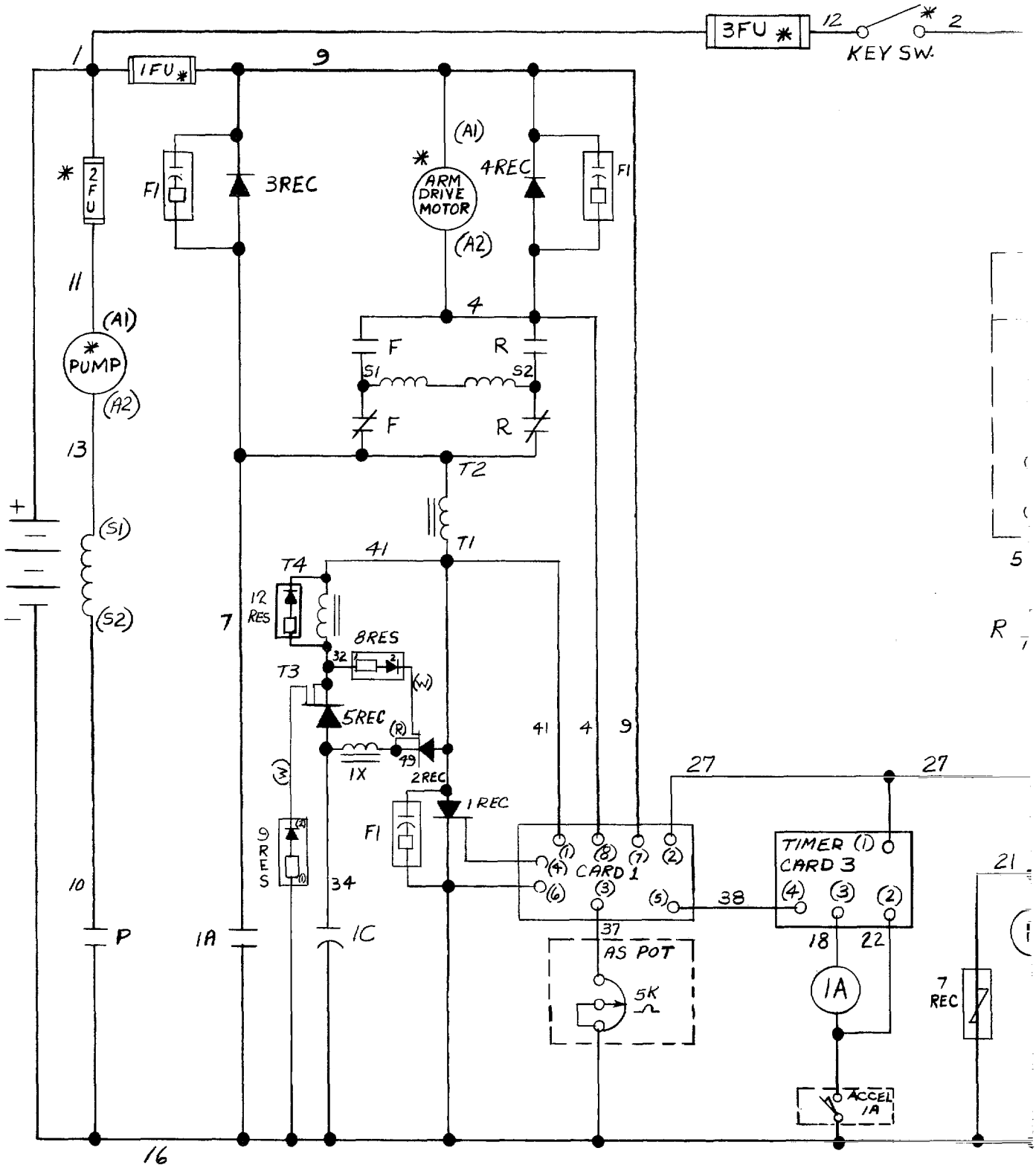


Figure 4—Typical elementary diagram, General Electric C-29

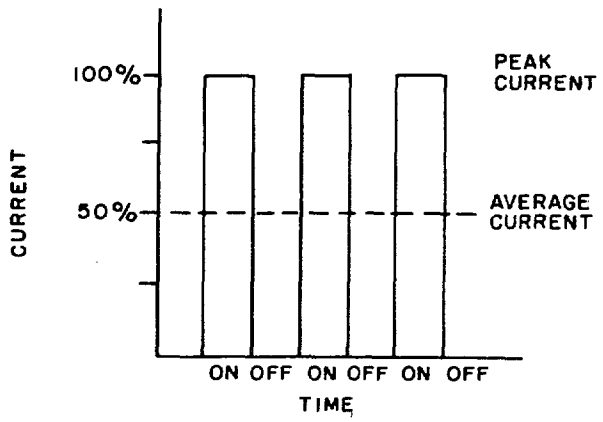


Figure 5—Battery Current

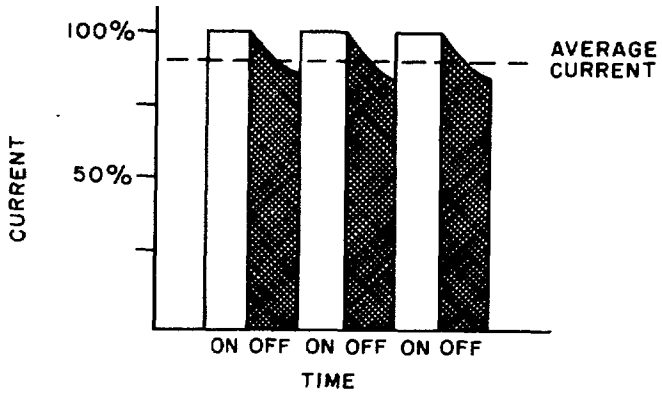


Figure 6—Motor Current

- CURRENT-LIMIT—The current-limit trimpot adjusts the frequency of the oscillator to limit the maximum SCR Current to 300A avg. (500-600A avg. Motor Current) at locked rotor. This circuit is sensitive to Load Current and overrides the oscillator under heavy load, thereby providing protection to motor and control by limiting currents during acceleration and stall.
- STATIC-PLUG—The static-plug trimpot adjusts the frequency of the oscillator to control the reverse torque applied to the motor to bring the vehicle to a safe smooth stop and/or reverse its direction.
- PROGRAMMED - ACCELERATION — The programmed-acceleration function of Card 1 has no external adjustment. With the accelerator in any position (from start to top speed) this circuit provides a gradual build-up of the pulse frequency over a given time, thereby preventing sudden high torque. This feature allows smooth acceleration from standstill to top SCR speed.

1A CONTACTOR (By-pass contactor around the SCR control)—The 1A contactor is used to provide top truck speed, torque, and efficiency when called for. The 1A contactor is picked up when the accelerator is moved to its extreme end of travel.

1A TIMER CARD (Card 3) — Time-delay acceleration is provided by a static timing circuit which delays the closing of the 1A contactor following the closing of the 1A switchette in the accelerator switch. This delay is adjustable by means of a trimpot on Card 3 and should be adjusted to give smooth acceleration on SCR control before bypassing.

THERMOSTAT (if used) — The thermostat is provided as a normally open temperature sensitive device to protect the SCR unit from overheating. It is mounted directly on the 1-REC heat sink and is connected between T1 and the accelerator potentiometer through a properly matched resistive bridge circuit. If during prolonged operation the 1-REC temperature tends to exceed the rated value, the thermostat will sense this condition and will close. Closing the thermostat inserts a fixed resistance in series with the accelerator potentiometer which reduces the frequency of the oscillator. This reduces the speed of the truck and the power flow through 1 REC. 1 REC will cool off and when 1 REC temperature drops to the preset value, the thermostat contacts will open and let the control resume its normal operation.

Other functions and equipment available with SCR control for electric vehicles and their instruction references are:

IC4484 FAULT DETECTOR .....(GEK-7940)

IC4484 PULSE MONITORING TRIP (GEK-28539)

IC4484Y100D STATIC RETURN-TO-OFF CARD ..... (GEK-7941)

IC4484 FIELD-WEAKENING SYSTEM (GEK-7942)

IC4484 IA HOLD-OFF CARD ..... (GEK-7972)

IC3012BH ACCELERATOR MASTER SWITCH .....(GEK-8073)

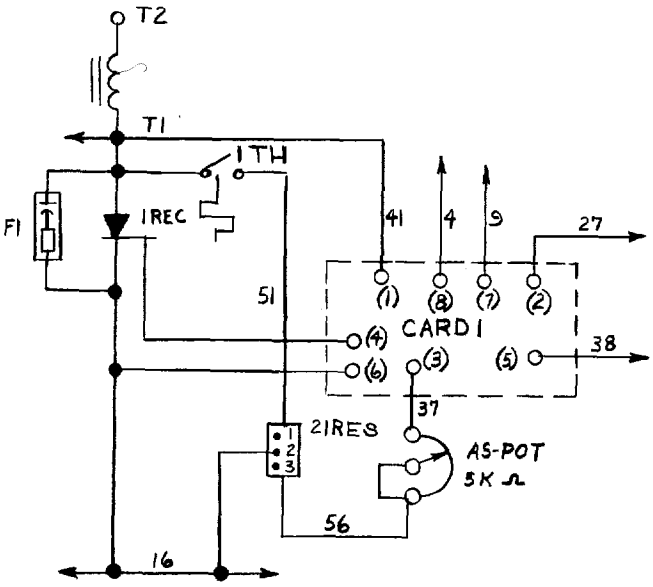


Figure 7—Partial elementary diagram showing thermostat connection

General Maintenance Instructions

The SCR control, like all electrical apparatus, does have some thermal losses. The semi-conductor junctions have finite temperature limits above which these devices may fail. For these reasons, normal maintenance should guard against any action which will expose the components to excessive heat, such as steam cleaning; or which will reduce the heat dissipating ability of the control, such as restricting air flow.

The following DO's and DON'Ts should be observed:

- Any controls that will be used in altitudes of 5000 feet or over, and in ambients of 100 degrees F or over should be brought to the attention of the truck manufacturer.
- The stall current should not be maintained for longer than 30 seconds.
- All external components having inductive coils must be filtered. Refer to vehicle manufacturer for specifications.
- The control should not be steam cleaned. Blowing off with a low-pressure air hose is the preferred method.
- The heatsinks of the SCR's and diodes are specially painted to help dissipate heat. Painting these heat sinks any other color will limit the ability of the sinks to dissipate heat.
- Terminal boards and other exposed SCR control parts should be kept free of dirt and paint which might change the effective resistance between points.
- The truck should not be plugged when the truck has been jacked up and the drive wheels are in a free wheeling position. This can create excessive voltages that can be harmful to the control.

Trouble-Shooting Instructions

The pulsing of the main SCR is too fast for conventional instruments to measure. When the control is functioning properly, a low hum can be heard.

Malfunctions of the SCR will generally fall into one of two categories. They are either no power (Table 1) or full power (Table 2), when operating in the SCR control range.

These simple and easy-to-follow tables outline the various symptoms and the corrective action to be taken.

The wire numbers and device descriptions are all referenced on pages 3, 4, 5, 13 and 14.

The same device designations have been maintained on different controls but the wire numbers may vary. Refer to the elementary and wiring diagram for your specific control. The wire numbers shown on the elementary diagram will have identical numbers on the corresponding wiring diagrams for a specific truck, but these numbers may be different from the numbers referenced in this publication.

Before proceeding, visually check for loose wiring, maladjusted linkage to accelerator switch, signs of overheating of components, etc. Before touching electrical components, disconnect the battery and discharge capacitor 1C. Reconnect the battery as needed for the specific check.

Tools and test equipment required are torque wrench, 18/36-volt test battery, 220-ohm 2-watt resistor, clip leads, volt-ohm meter (20,000 ohms per volt) and general hand tools.

Table 1

Failures Which Cause No Motor Torque With SCR Control

	SYMPTOMS	WHAT TO DO
1A.	Zero volts from control positive to control negative.	● Check 1 FU and 3 FU to see if they are blown. ● Check the key switch for misoperation. ● Check battery for low specific gravity and connections for looseness or broken fittings.
1B.	Contactors (F or R) do not pick up. Battery volts from control positive to control negative.	● The problem may be an open circuit from control positive to F or R coils, including misoperation of the seat switch, brake switch, start switch, directional switch or normally-closed F or R interlocks. ● Check 7 and 8 REC for short circuits (See Table 4-F).
1C.	Contactors close (F or R). No power and no SCR hum with accelerator in SCR range.	● With the accelerator in operating range, 1A blocked open, and contactors F or R picked up, check for control volts between terminal 2 Card 1 and negative. If there is zero volts at this point, check F and R interlocks for open circuit. ● Check for battery volts between terminals (1) and (6) on Card 1. If voltage is not present, check the power contact tips of F and R and the continuity of wire 41 from T1 to Card 1 (1) and from Card 1 (6) to battery negative. ● Check for about one third of control volts between Card 1 terminals 3 and 6 (except near high-speed position of accelerator). If near zero or about half or more of control volts (except near high-speed position of accelerator when the volts will rise to control volts), the potentiometer (and/or contacts) in the accelerator may be defective (See Table 4-I), or Card 1 may be defective (See Table 4-A). ● Check for open circuit from Card 1 terminal 4 thru 1 REC gate (white lead). (See Table 4-G). ● Check 1 REC for open gate (See Table 4-G). ● Check oscillator section of Card 1 (See Table 4-A).
1D.	Contactors close (F or R). Very little power and high-pitch SCR hum.	● Check 2 REC for a short in the conducting direction (See Table 4-G).
1E.	Contactors close (F or R). Normal hum but little or no torque.	● Check for short in 3 and 4 REC (See Table 4-E).

Table 2

Failures Which Cause Full Motor Torque With SCR Control

	SYMPTOMS	WHAT TO DO
2A.	Contactors close (F or R). Full SCR speed immediately. Audible hum.	● Check for short from negative to terminal 3 of Card 1. ● Check potentiometer for proper resistance (See Table 4I).
2B.	Contactors close (F or R). Full speed immediately. No audible hum.	● Check for 1A contactor failure to open, or short in truck wiring, or malfunction of Card 3 causing 1A to pick up immediately. (See Table 4-B).
2C.	Contactors close (F or R). Full speed immediately. No audible hum. Capacitor 1C not charged when measured within 30 minutes after operation of truck.	● Check 1 REC for a short (See Table 4-G). ● Check for open in 5 REC gate circuit from negative through 9 RES to 5 REC. ● Check 9 RES (See Table 4-H). ● Check 5 REC for open gate (See Table 4-G). ● Check continuity of wires from 1C to 5 REC and from 5 REC through T3, T4 to T1 at the 1 REC anode. ● Check for shorts in capacitor 1C (See Table 4-C).
2D.	Contactors close (F or R). Full speed immediately. No audible hum. Capacitor 1C charged, i.e., volts from battery negative (positive on voltmeter) to 5 REC anode wire 34 is more than 30 volts.	● Check for open in 2 REC gate circuit from T3 through 8 RES to 2 REC. ● Check 8 RES (See Table 4-H). ● Check 2 REC (See Table 4-G).

Table 3

Misoperation of Special Features

	SYMPTOMS	WHAT TO DO
3A.	Failure of 1A contactor to operate.	● Check voltage between terminals 1 and 2 of Card 3 (When F contactor is closed) to be control volts. If voltage is not present check for open in wiring from positive to Card 3 terminal 1. Also check for open from terminal 2 - Card 3 through 1A switch on accelerator, to control negative. ● Check voltage between terminals 3 and 2 of Card 3 to be battery volts after time delay. If voltage is present, check for open circuit in the 1A coil. (See Table 4D). ● Check Card 3 (See Table 4-B).

	SYMPTOMS	WHAT TO DO
3B.	Failure of FW contactor to operate (if used).	● Refer to GEK-7942.
3C.	Failure of Static Return-to-off circuit (if used).	● Refer to GEK-7941.
3D.	Failure of 1A Hold-off circuit (if used).	● Refer to GEK-7972.
3E.	Failure of Fault-detector circuit (if used).	● Refer to GEK-7940.
3F.	Severe reversal.	● Adjust static-plug trimpot on Card 1, turning the pot counterclockwise for softer plug. ● Check 4 REC (See Table 4-E). ● Check current limit (See Table 5).
3G.	Severe reversal. (Deceleration smooth but severe as vehicle changes direction).	● Check adjustment and operation of timer card. (See Table 4B).
3H.	Reversal too soft.	● Check static-plug trimpot on Card 1 (Adj. clockwise to stiffen plug).

Table 4

Checking Components

4A. CARD 1
Tune up card per Table 6. If these adjustments cannot be made or if control fails to operate satisfactorily, check all other components. If all check good, replace card with a known operating card.

4B. CARD 3 (1A Timer)
Disconnect leads to Card 3. Connect a voltmeter (50V scale on 36 volt battery or 250V scale on 72 volt battery) from terminal 3 to 2. Apply battery volts negative to terminal 2 and positive to terminal 1. The voltmeter should read about one volt and then raise to battery volts at the end of the timer interval, .3 to 2.5 seconds. Turning trimpot should adjust time (CW to increase). Remove battery volts and voltmeter should drop to zero.

4C. CAPACITOR 1C
Disconnect battery and discharge capacitor. Remove wire No. 34 from all capacitors. Measure ohms through each capacitor using the RX10,000 scale. Meter should read zero ohms and then swing to above 100,000 ohms. If not, replace capacitors.

4D. CONTACTORS (F, R, 1A, FW AND P)
a) 100-ampere contactors. (See instructions GEH 3086.)
b) 200-ampere contactors. (See instructions GEH-3074.)

NOTE: Control is arranged so that F and R do not break current. Contactor 1A drops out ahead of F or R.

4E. RECTIFIERS

When checking diodes, *disconnect battery and discharge capacitor 1C* to prevent burning out the ohmmeter. When reassembling hardware on rectifiers, refer to Table 5 for proper torque settings.

To check 3, 4, 15 and 16 REC, disconnect one side of the rectifier. These are diodes with about 7 to 12 ohms in the conducting direction ($\overset{+}{\longrightarrow}\!\!\!\!\!\!\!\!\!\!\!\!\overset{-}{}$) measured with scale RX1, and 50,000 ohms or higher in the non-conducting direction ($\overset{-}{\longrightarrow}\!\!\!\!\!\!\!\!\!\!\!\!\overset{+}{}$) measured with scale RX10,000. (On some meters about 75 to 160 ohms are obtained in the conducting direction.)

4F. CONTACTOR COIL AND ACCESSORY FILTERS
(7, 8, & 9 REC) (Typical Cat. No. 148B6203G14)
These are varistors and should be checked as follows: Disconnect battery and discharge capacitor 1C. Disconnect the leads to the filter block. Connect a 36-volt d-c test battery in series with the varistor and a volt-ohmmeter set on the 1 ma. scale as shown in Figure 8. If the varistor is good, there will be a noticeable deflection of the meter needle when the leads are touched to the filter block terminals. If no deflection is obtained replace the filter block.

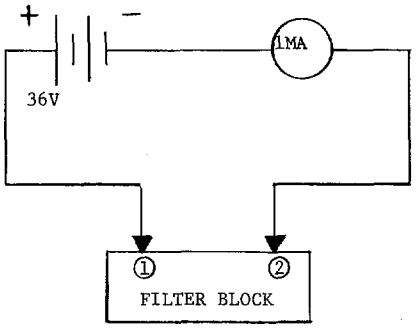


Fig. 8

4G. SCR's (1 REC, 2 REC and 5 REC)
These are Silicon Controlled Rectifiers. Before checking, disconnect battery and discharge capacitor 1C to prevent burning out the ohmmeter.

a) 1 REC—To check 1 REC, disconnect the heavy pigtail lead and the small gate lead. Measure resistance from stud to heavy lead then reverse polarity of the meter and repeat above step. The resistance in both directions should be infinite (RX10,000 scale). Replace 1 REC if the resistance is below 100,000 ohms. Measure resistance between the heavy pigtail lead and the small lead (gate). With ohmmeter positive on the gate lead, the resistance should measure between 10 and 150 ohms on the RX1 scale. Now reverse the ohmmeter leads; resistance should be 20 ohms or above on the RX1 scale.

b) 2 REC and 5 REC—To check either REC, disconnect heavy pigtail lead and small white lead (also disconnect small red lead, if present). Measure resistance from stud to heavy lead, and then reverse polarity on the ohmmeter and repeat above step. Resistance should be infinite in both directions (RX10,000 scale). Replace REC if resistance is below 100,000 ohms. Measure resistance between the heavy pigtail lead and the small white lead (gate). With ohmmeter positive on the white lead, the resistance should measure between 20 and about 300 ohms on the RX1 scale. Now reverse the ohmmeter leads; resistance should be 100 ohms or above on the RX1 scale.

NOTE: SCR's may check outside the above limits and still operate satisfactorily. In all cases the measured resistance with ohmmeter positive lead on the gate lead will be lower than when this lead is reversed and placed on the pigtail.

4H. GATE CONTROL (8 RES, 9 RES)
8 and 9 RES act like a diode with high resistance in the conducting direction.
To check, disconnect the lead from terminal 2 to the gate. Using a test battery, 220-ohm resistor and voltmeter as in Figure 9; the voltmeter should read about 22 volts when the voltmeter is connected to terminal 1 (positive) and terminal 2 (negative). Reverse voltmeter leads (terminal 2 negative), voltmeter should read battery volts.

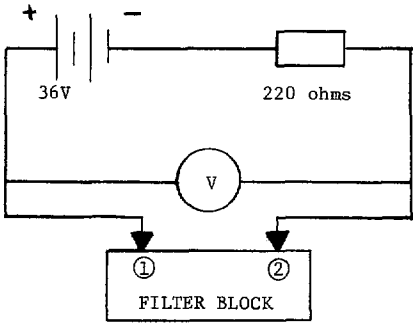


Fig. 9

4I. ACCELERATOR POTENTIOMETER—
Check procedure: Disconnect pot from circuit by removing wire 37 from card 1 terminal 3. Measure resistance of pot on RX100 scale between wire 37 and negative wire 16 (wire 56 terminal 3 of 21 RES if thermostat is used—refer to Figure 7). Resistance should read approximately 5000 ohms and smoothly reduce to zero as the foot-pedal is depressed to maximum travel.
Adjustment: To set potentiometer resistance, as the foot-pedal is depressed from off towards maximum travel, the resistance should be 3500 ohms or higher as the start switchette operates. As you continue to depress the pedal the resistance should be less than 300 ohms when the 1A switchette operates. If the resistance does not read as above, loosen the screw clamping the gear to the pot shaft and adjust pot shaft position until above reading is obtained.

4J. SEMI-CONDUCTOR FILTER BLOCKS (F1)
To check, disconnect all wires from filter block. With the VOM on RX10,000 scale—touch the leads to terminals 1 and 5 to charge the filter. Reverse the meter leads touching them to terminals 5 and 1. A strong capacitor action should be observed (i.e., the VOM needle will deflect and return to infinity). This same check must be applied to terminals 2 and 6, and 3 and 7. If this capacitor action is not observed, replace filter block. NOTE: There is no filter between terminals 4 and 8.

4K. TRANSFORMER SECONDARY FILTER (12 RES)
12 RES acts like a diode with high resistance in the conducting direction. To check, disconnect the leads from terminals 1 and 2. Using a test battery, 220-ohm resistor and voltmeter as in 4H, the voltmeter should read about 31 volts when the voltmeter is connected to terminal 1 (positive) and terminal 2 (negative). Reverse voltmeter leads (terminal 2 negative), voltmeter should read battery volts.

4L. SEMI-CONDUCTOR FILTER BLOCK (SF)
To check, disconnect all wires from filter block. With VOM on RX10,000 scale, touch the leads to the filter terminals to charge the filter. After a few seconds, reverse the meter leads and touch the filter terminals. The VOM needle will deflect and return to infinity. If this capacitor action is not observed, replace the filter block.

Table 5

Checking Current-Limit

5A. The current-limit circuit has been preset at the factory and should not be adjusted. However, if truck operation appears sluggish or if control appears to be overheated, the following check should be made.

Material required to check current-limit:

500-ampere 50-millivolt d-c shunt

50-millivolt d-c meter (d'Arsonval movement)

50-volt d-c meter (d'Arsonval movement) (250-volt scale needed for 72V)

Battery with equal or greater ampere-hour capacity used on truck, charged to 1200 or higher specific gravity.

Check that ohms in accelerator potentiometer are under 300 ohms in high speed (See Table 4-I).

Connect the shunt and millivolt meter between battery negative and 1 REC. Connect the voltmeter across the motor (between battery positive and T2 on the SCR panel). Disconnect the positive lead of 1A contactor coil or block power tips of 1A contactor open. Jack the wheels of the truck off the floor. If a brake interlock is used, short it out so that power and brakes can be applied at the same time.

Connect battery and depress the accelerator slightly to assure that a positive deflection of the meters are obtained.

Check maximum speed adjustment first by moving the accelerator to the full-power position and applying the brakes until battery current is about 70-80 percent of the loaded level amperes (approximately 100 amps) with motor directly across the battery and motor fields connected the same as are used in SCR control.

Voltmeter should read between the following:

Battery Volts	*Motor Volts*
36	25-30
48	35-40
72	50-60

Now, check current-limit by moving the accelerator to the full-power position and applying the brakes until the wheels come to a standstill or until the meter reads 300 amperes, plus or minus 15 amperes (full scale is 500 amperes).

NOTE: Do not stall the motor for more than 30 seconds at a time. Allow time for motor cooling between stalls. Do not operate motor at high speeds or plug the motor with wheels jacked up.

5B. If the current-limit is not properly set, refer to the truck manufacturer. If either adjustment has to be altered, recheck maximum speed and current-limit until both limits are met. The maximum speed adjustment has a slight effect on current-limit. The current-limit adjustment has a slight effect on high speed. Both adjustments are CW to increase.

5C. Torques for Rectifiers*

a)	1 REC (C290)	300 inch-pounds, or	24 foot-pounds
b)	2 and 5 REC (C50)	150 inch-pounds, or	12 foot-pounds
c)	3 and 4 REC	300 inch-pounds, or	24 foot-pounds

* Values given are nominal, limits are plus-or-minus 50%.

NOTE: Silicon grease (Penatrox A) should always be applied between rectifier and heat-sink before tightening rectifier hardware.

Table 6

Tuneup for New or Mistuned Card 1

Panels are factory adjusted for a particular motor and truck and should not need adjustment when used with this motor and truck. If minor adjustments are required for optimum performance they may be made without regard to the following procedure because the adjustments are so designed that they do not interact when near their proper setting.

If the panels are used to control motors or trucks for which they were not factory adjusted, the settings may be out of optimum adjustment to the extent that they do interact and the following procedure must be followed:

(1) Turn current-limit adjustment fully clockwise.
(2) Turn static-plugging adjustment fully clockwise.
These two steps prevent any interaction when setting the speed adjustments.
(3) Adjust creep-speed as desired.
(4) Adjust top SCR speed per Table 5 and with 1A power tips blocked open.
(5) Turn current-limit and static-plug trimpots fully counterclockwise and then back CW about 5 turns. NOTE: More than 5 turns may be required before control will operate.
(6) Set current-limit for desired battery current with the motor in stall condition per Table 5.
(7) With truck on the ground set static-plugging for desired plugging distance.

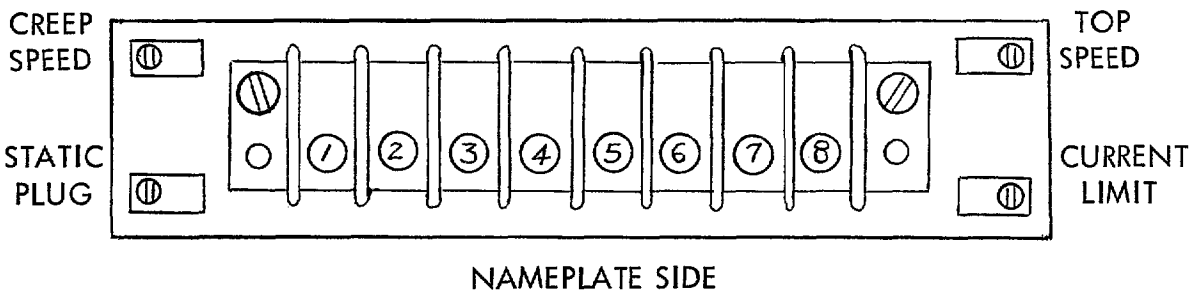
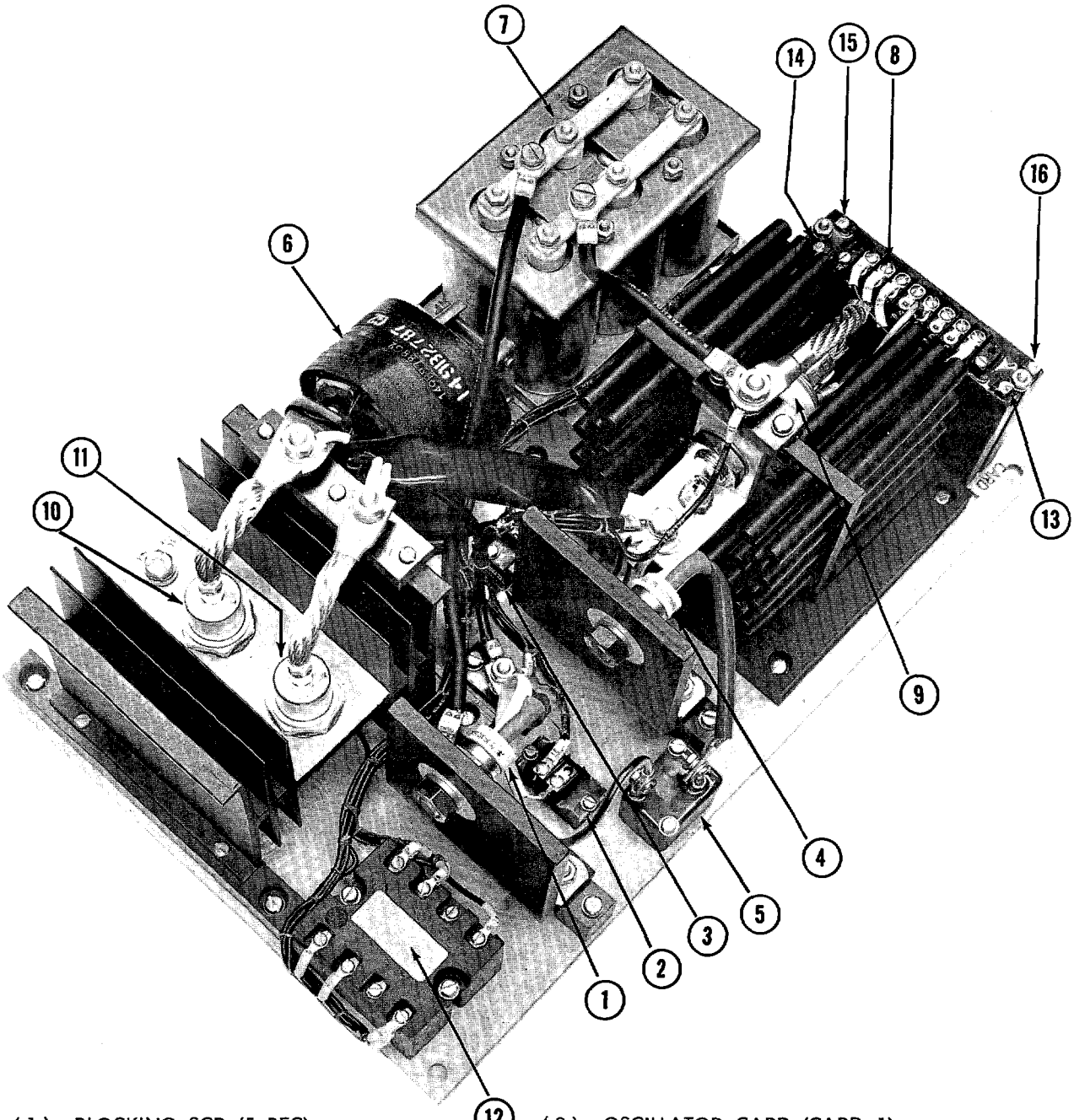


Figure 10—Card 1 Trimpot Adjustments.

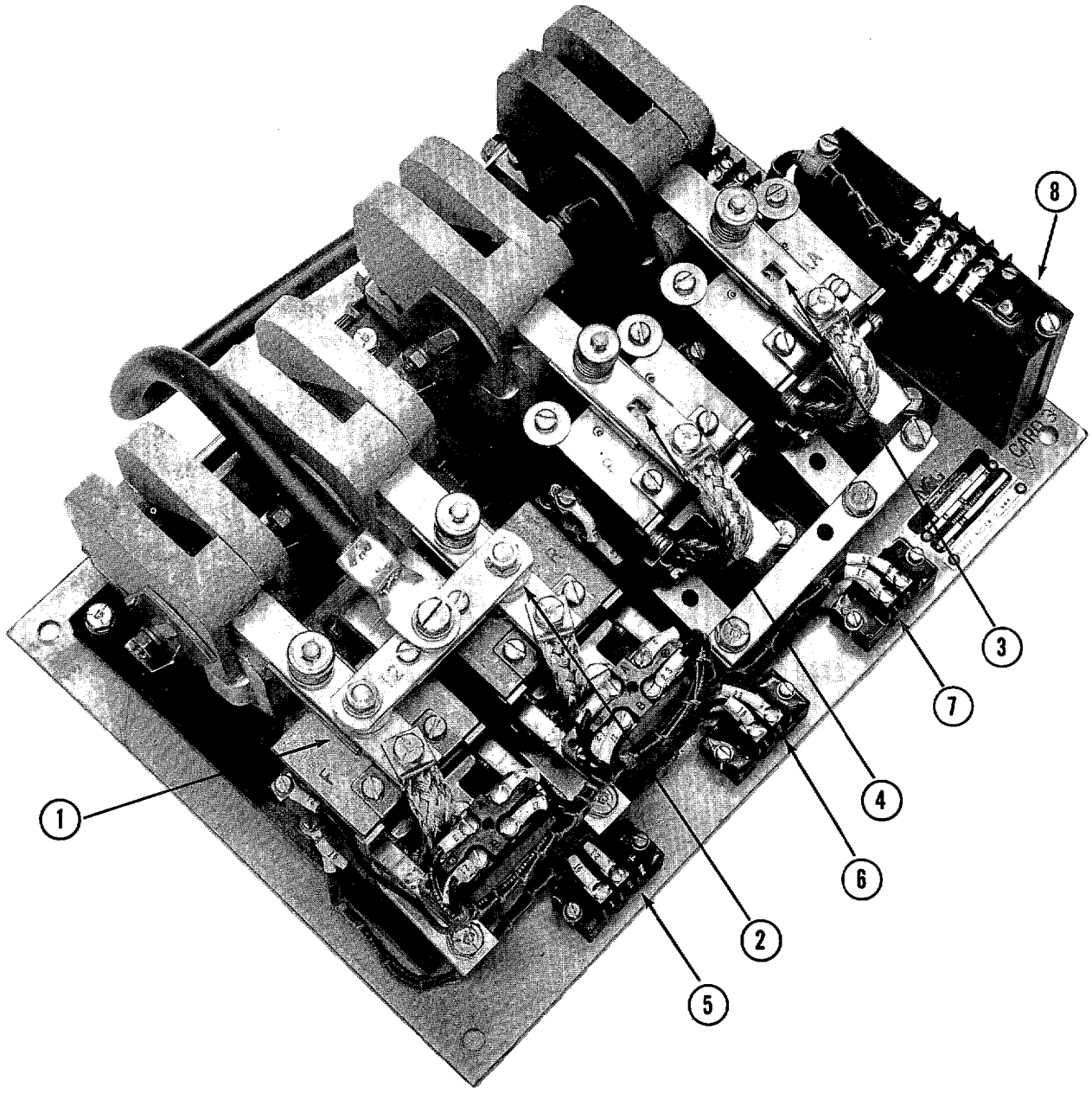
TYPICAL PHYSICAL ARRANGEMENT AND IDENTIFICATION OF COMPONENTS

(Refer to wiring diagram furnished with truck for precise arrangement of components.)



(1) BLOCKING SCR (5 REC)
(2) GATE CONTROL FOR BLOCKING SCR (9 RES)
(3) GATE CONTROL FOR TURN-OFF SCR (8 RES)
(4) TURN-OFF SCR (2 REC)
(5) CURRENT-LIMITING REACTOR (1X)
(6) PULSE TRANSFORMER (1T)
(7) COMMUTATING CAPACITOR (1C)
(8) OSCILLATOR CARD (CARD 1)
(9) MAIN SCR (1 REC)
(10) FLY-BACK DIODE (3 REC)
(11) PLUGGING DIODE (4 REC)
(12) FILTER BLOCK (1F for 1, 3 and 4 REC)
(13) CREEP-SPEED ADJUSTMENT
(14) TOP-SPEED ADJUSTMENT
(15) CURRENT-LIMIT ADJUSTMENT
(16) STATIC-PLUGGING ADJUSTMENT

Figure 11—Typical C-290 Static Panel.



(1) FORWARD CONTACTOR (F)
(2) REVERSE CONTACTOR (R)
(3) BY-PASS CONTACTOR (1A)
(4) PUMP CONTACTOR (P)
(5) COIL SUPPRESSOR (7 REC)
(6) COIL SUPPRESSOR (8 REC)
(7) COIL SUPPRESSOR (9 REC)
(8) STATIC-TIMER CARD (CARD 3)

Figure 12—Typical Magnetic Panel.

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